Discussion

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Summary

CFM in Zambia is broadly working but there is significant room for improvement. On the positive side, a clear majority of communities report that CFM has reduced deforestation, lowered fire frequency, and increased wildlife populations — the core indicators of SFM. Most communities also recognise livelihood gains from increased employment and income, and enumerator observations confirm that many CFMGs are actively engaged and functioning.

However, governance remains a critical weakness. Many CFMGs have not completed basic legal and administrative processes, financial management processes are often absent, and DFO oversight is infrequent despite the statutory role of the FD. A number of CFMGs are dormant or struggling, and community feedback is highlights many failings, especially around transparency and benefit sharing. The enumerator observations, which flagged capacity gaps, re-engagement needs, dormant CFMGs, leadership challenges, and elite capture, paint a picture of CFM implementation that is on the verge of failing in a significant minority of cases.

Economic transformation is limited and over-reliant on a single revenue stream, namely carbon. At the time of the survey, on around one third of CFMGs had received carbon payments. Honey was the only other forest commodity that was being widely exploited, but only by about 20% of CFMGs. The broader potential of forest product enterprises — documented across a wide diversity of NTFPs — remains largely untapped.

Equity outcomes are broadly positive, with CFM appearing to give women a formal voice in forest management they often lacked before. However, youth marginalisation and elite capture are visible in a minority of CFMGs and must be actively identified and addressed.

Communities themselves have a clear understanding of what is and is not working, and their feedback on training needs, monitoring approaches, and areas for improvement, as documented in this chapter, provides a practical roadmap for strengthening CFM across Zambia.

Feedback from the communities

What is working in CFM

We asked the community FGD groups to identify aspects of CFM that they feel are working well in their communities (Figure 1). The most frequently mentioned reflect different aspects of SFM. Clearly, most communities do feel that CFM has led to improvements in forest management, reducing deforestation and enhancing wildlife conservation. Also mentioned, but less frequently, were indicators of good governance and improved livelihoods. Evidently, CFM is working well for some communities.

```
prefix_w <- "opportunitiesb_for_improved_cfm_implementation_cfm_working_aspects_"

working_labels <- c(
  "effective_management_of_forest_resources_e_g_sustainable_harvesting_restoration" = "Forest
  "reduction_in_deforestation_and_forest_degradation"                               = "Reduce
  "increased_wildlife_conservation_and_biodiversity"                                = "Wildli
  "equitable_benefit_sharing_among_community_members"                               = "Equita
  "active_participation_of_women_youth_and_marginalized_groups_in_cfm_activities"  = "Inclusi
  "improved_livelihoods_and_income_generating_opportunities_through_cfm_activities" = "Improv
  "successful_partnerships_with_private_enterprises_or_external_institutions"       = "Partne
  "successful_partnerships_with_traditional_leaders"                                = "Tradit
  "improved_infrastructure"                                                          = "Impro
  "effective_monitoring_and_reporting_of_cfm_outcomes_and_impacts"                 = "Effecti
  "transparent_and_fair_decision_making_processes_within_the_community"            = "Transpa
  "strong_community_awareness_and_involvement_in_cfm_activities"                   = "Communi
)

full_w_vars <- paste0(prefix_w, names(working_labels))

# Frequency counts from binary checkbox variables
working_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_w_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = working_labels[sub(prefix_w, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)

# Word frequencies from open-ended comments
comment_freq <- cfmg |>
```

```
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(comment = opportunitiesb_for_improved_cfm_implementation_cfm_working_aspects_comment
  drop_na(comment) |>
  unnest_tokens(word, comment) |>
  anti_join(stop_words, by = "word") |>
  filter(nchar(word) > 2, !word %in% c("cfm", "cfmg", "forest", "community", "people")) |>
  count(word, name = "freq") |>
  filter(freq > 1)

# Combine checkbox labels and comment words
working_all <- bind_rows(working_freq, comment_freq) |>
  group_by(word) |>
  summarise(freq = sum(freq), .groups = "drop")

set.seed(42)
ggplot(working_all, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 22)) +
  scale_colour_gradient(low = "#6ABFB8", high = "#1A6B63") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 1: Aspects of CFM that are working well, as identified by community female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each aspect.

Areas for improvement

We also asked the community FGD groups to identify the aspects of CFM that they feel need improvement in their communities (Figure 2). The diversity of responses was greater and most can be taken as indicators of poor governance, such as 'community awareness', 'transparent decision-making' and 'inclusive participation'. Economic indicators also featured prominently, such as 'improved livelihoods' and 'partnerships'. Indicators of SFM were also mentioned indicating that some communities feel there is scope for improvement there too. Clearly, there is room for improvement in CFM, especially in governance aspects.

```
prefix_i <- "opportunitiesb_for_improved_cfm_implementation_cfm_areas_for_improvement_"

improvement_labels <- c(
  "effective_management_of_forest_resources"                       = "Forest management",
```

```
  "reduction_in_deforestation_and_forest_degradation"             = "Reduced deforestation",
  "increased_wildlife_conservation_and_biodiversity"              = "Wildlife conservation",
  "equitable_benefit_sharing_among_community_members"             = "Equitable benefits",
  "active_participation_of_women_youth_and_marginalized_groups"   = "Inclusive participation"
  "improved_livelihoods_and_income_generating_opportunities"      = "Improved livelihoods",
  "partnerships_with_private_enterprises_or_external_institutions" = "Partnerships",
  "effective_monitoring_and_reporting_of_cfm_outcomes"            = "Effective monitoring",
  "transparent_and_fair_decision_making_processes"                = "Transparent decision mak
  "community_awareness_and_involvement_in_cfm_activities"         = "Community awareness",
  "capacity_building"                                             = "Capacity building",
  "infrastructure"                                                = "Infrastructure",
  "alternative_livelihoods"                                       = "Alternative livelihoods"
  "transport"                                                     = "Transport",
  "governanace_leadership_institutional_strengthening"            = "Governance & leadership"
  "external_funding"                                              = "External funding",
  "improved_water_security"                                       = "Water security",
  "human_wildlife_conflict"                                       = "Human-wildlife conflict"
)

full_i_vars <- paste0(prefix_i, names(improvement_labels))

# Frequency counts from binary checkbox variables
improvement_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_i_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = improvement_labels[sub(prefix_i, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)

# Word frequencies from open-ended comments
improvement_comment_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(comment = opportunitiesb_for_improved_cfm_implementation_cfm_areas_for_improvement_
  drop_na(comment) |>
  unnest_tokens(word, comment) |>
  anti_join(stop_words, by = "word") |>
  mutate(word = case_match(word, "hfos" ~ "HFOs", .default = word)) |>
  filter(nchar(word) > 2,
         !word %in% c("cfm", "cfmg", "forest", "community", "people",
                      "products", "visits", "water", "activities", "provide",
```

```
                      "visitation", "production", "increase", "improved",
                      "regular", "change", "continue")) |>
  count(word, name = "freq") |>
  filter(freq > 1)

# Combine checkbox labels and comment words
improvement_all <- bind_rows(improvement_freq, improvement_comment_freq) |>
  group_by(word) |>
  summarise(freq = sum(freq), .groups = "drop")

set.seed(42)
ggplot(improvement_all, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 22)) +
  scale_colour_gradient(low = "#F4A582", high = "#8B0000") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 2: Aspects of CFM identified as needing improvement, as identified by community female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each aspect.

External challenges

We asked the community FGD groups to identify the external challenges facing CFM in their communities (Figure 3). Perhaps unsurprisingly, the responses focus on development issues, such as 'financial constraints', 'poor infrastructure', 'limited market access' and 'insufficient technical expertise'.

```
prefix_ec <- "opportunitiesb_for_improved_cfm_implementation_cfm_external_challenges_"

external_challenge_labels <- c(
  "government_policies_and_regulations"                  = "Policy/regulatory barriers",
  "lack_of_external_funding"                             = "Financial constraints",
  "limited_market_access_for_forest_products"            = "Limited market access",
  "conflicts_with_neighboring_communities"               = "Conflicts with neighbours",
```

```
  "conflicts_within_community"                                 = "Internal conflicts",
  "conflicts_with_neighboring_with_partners"                   = "Conflicts with partners",
  "insufficient_technical_expertise"                           = "Insufficient technical expert
  "weak_enforcement_of_laws_and_regulations"                   = "Weak law enforcement",
  "political_instability_or_government_changes"                = "Political instability",
  "poor_infrastructure"                                        = "Poor infrastructure",
  "natural_disasters"                                          = "Natural disasters",
  "limited_support_from_govt"                                  = "Limited government support",
  "limited_support_from_tradtional_leaders"                    = "Traditional leader challenges
  "water_social_services_and_basic_needs"                      = "Limited government support",
  "human_wildlife_conflicts"                                   = "Human-wildlife conflict",
  "transport_and_communication"                                = "Transport/communication",
  "chiefdom_succession_disputes"                               = "Chiefdom succession disputes'
  "awareness_information_and_capacity_gaps"                    = "Awareness/capacity gaps",
  "cultural_or_social_resistance_to_cfm_initiatives"          = "Cultural/social resistance",
  "traditional_leadership_and_governance_interference"         = "Traditional leader challenges
  "social_economic_and_environmnetal_stressors"                = "Socioeconomic stressors",
  "policy_and_regulatory_barriers"                             = "Policy/regulatory barriers",
  "enchroachment_illegal_activities_and_extternal_pressure"    = "Encroachment/illegal activit
  "partner_related_challenges_and_carbon_market_issues"        = "Partner/carbon market issues'
  "financial_material_and_logistical_constraints"              = "Financial constraints"
)

full_ec_vars <- paste0(prefix_ec, names(external_challenge_labels))

# Frequency counts from binary checkbox variables
external_challenge_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_ec_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = external_challenge_labels[sub(prefix_ec, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)

external_challenge_all <- external_challenge_freq |>
  group_by(word) |>
  summarise(freq = sum(freq), .groups = "drop")

set.seed(42)
ggplot(external_challenge_all, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
```

```
scale_size(range = c(3, 22)) +
scale_colour_gradient(low = "#F4A582", high = "#8B0000") +
theme_minimal(base_size = 14) +
theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 3: External challenges facing CFM, as identified by community female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each challenge.

Monitoring methods

We asked the FGD groups to identify the methods they recommend are used to monitor CFM progress and outcomes in their communities (Figure 4). While a large number of approaches were mentioned, the community groups overwhelmingly identified community meetings and field visits/patrols as being the preferred approaches for monitoring CFM outcomes. However, it is interesting to note that female groups were far more likely to mention community meetings and much less likely to mention field visits/patrols, than the male and youth groups. This likely reflects the gender biases in participation in field visits and patrolling, which from earlier data are almost exclusively a male prerogative.

```
prefix_m <- "opportunitiesb_for_improved_cfm_implementation_cfm_monitoring_methods_"

monitoring_labels <- c(
  "community_meetings_to_review_progress_and_challenges"                                  = "
  "field_visits_or_patrols_to_assess_forest_conditions_and_resource_use"                 = "I
  "data_collection_and_surveys_on_forest_health_biodiversity_livelihoods"                = "I
  "financial_reporting_and_audits_to_track_resource_allocation_and_benefits"             = "I
  "independent_evaluations_by_external_bodies_ngo_s_government_agencies"                 = "I
  "participatory_monitoring_involving_community_members_in_data_collection_and_analysis" = "I
  "use_of_technology_or_tools_satellite_images_gis_mobile_apps"                          = "T
  "feedback_or_grievance_mechanisms_to_capture_community_concerns"                       = "I
)

full_m_vars <- paste0(prefix_m, names(monitoring_labels))

cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  mutate(group = factor(
    case_match(
      interview_type,
      "cfmg_female" ~ "Female",
      "cfmg_male"   ~ "Male",
      "cfmg_youth"  ~ "Youth"
    ),
    levels = c("Youth", "Male", "Female")
  )) |>
  select(group, all_of(full_m_vars)) |>
  pivot_longer(all_of(full_m_vars), names_to = "variable", values_to = "val") |>
  filter(val == 1) |>
  mutate(method = factor(
    monitoring_labels[sub(prefix_m, "", variable, fixed = TRUE)],
    levels = rev(monitoring_labels)
```

```r
)) |>
count(group, method) |>
complete(group, method, fill = list(n = 0)) |>
ggplot(aes(y = method, x = n, fill = group)) +
geom_col(position = position_dodge(width = 0.75), width = 0.65, colour = "white") +
geom_text(
  aes(label = ifelse(n > 0, n, "")),
  position = position_dodge(width = 0.75),
  hjust = -0.3, size = 3.5, colour = "grey30"
) +
scale_fill_manual(
  values = c(
    "Female" = "#2A9D8F",
    "Male"   = "#57B8B0",
    "Youth"  = "#E9C46A"
  ),
  guide = guide_legend(reverse = TRUE)
) +
scale_x_continuous(
  expand = expansion(mult = c(0, 0.15)),
  name   = "Number of FGDs"
) +
scale_y_discrete(labels = \(x) str_wrap(x, width = 25), expand = expansion(add = 0.6)) +
labs(y = NULL, fill = NULL) +
theme_minimal(base_size = 14) +
theme(
  panel.grid.major.y = element_blank(),
  panel.grid.minor   = element_blank(),
  panel.grid.major.x = element_blank(),
  axis.line.x.bottom = element_line(colour = "grey30", linewidth = 0.5),
  axis.text.x        = element_text(colour = "grey20"),
  axis.text.y        = element_text(size = 14, colour = "grey20"),
  legend.position    = "bottom",
  plot.margin        = margin(10, 25, 10, 10)
)
```

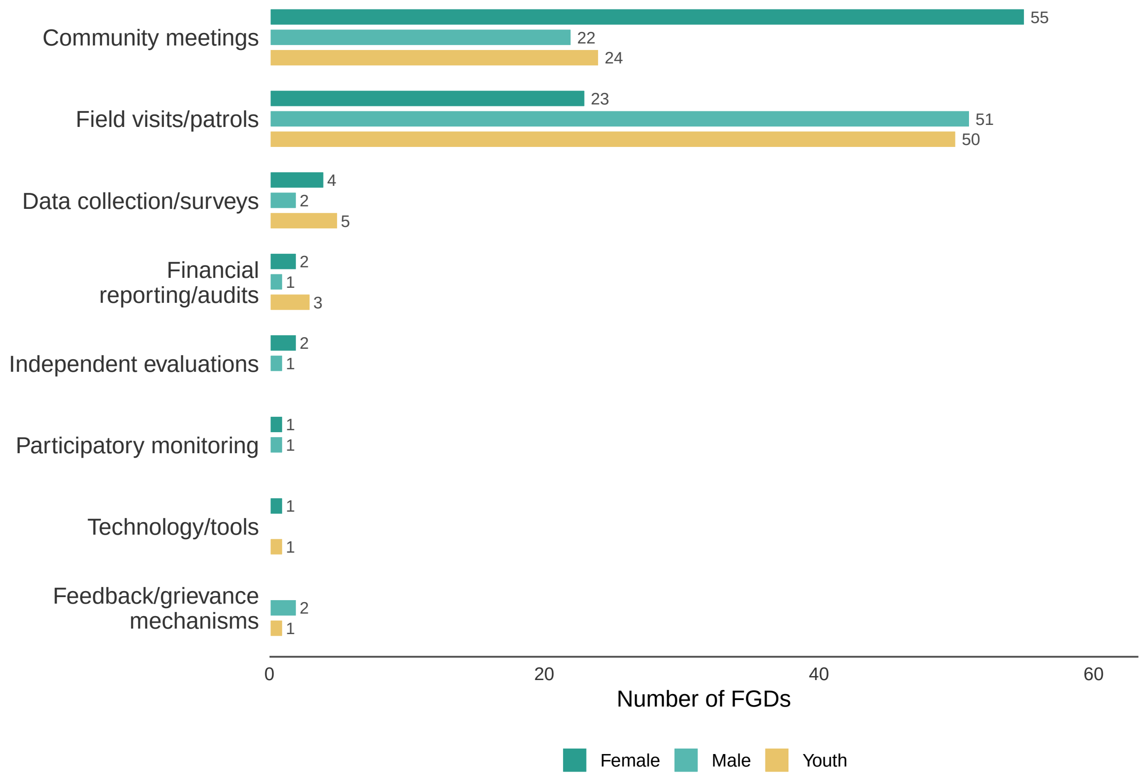

Figure 4: Methods used to monitor CFM progress and outcomes, as reported by community female, male, and youth FGDs. Bars show the number of FGDs reporting each method.

We also asked the community FGD groups how monitoring could be improved (Figure 5). Unsurprisingly, increased funding and more training featured prominently. However, it was interesting to note that digital tools and community feedback sessions were also mentioned frequently.

```
prefix_mi <- "opportunitiesb_for_improved_cfm_implementation_cfm_monitoring_improvement_"

monitoring_imp_labels <- c(
  "provide_more_training_for_cfm_group_members_on_monitoring_techniques"                =
  "introduce_digital_tools_for_real_time_data_collection_and_reporting"                 =
  "increase_funding_for_monitoring_activities"                                           =
  "establish_regular_community_feedback_sessions_to_discuss_monitoring_results"         =
  "develop_standardized_reporting_templates_for_consistent_data_collection"             =
  "improve_transparency_by_sharing_monitoring_reports_with_the_community"               =
```

```
  "increase_participation_of_women_youth_and_marginalized_groups_in_monitoring_activities" =
  "increase_hfos_scouts_guards"                                                             =
  "improve_transport_communication"                                                         =
  "ppe"                                                                                     =
  "increase_patrols"                                                                        =
  "improve_security_for_hfos"                                                               =
  "collaboration"                                                                           =
)

full_mi_vars <- paste0(prefix_mi, names(monitoring_imp_labels))

# Frequency counts from binary checkbox variables
monitoring_imp_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_mi_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = monitoring_imp_labels[sub(prefix_mi, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)

# Word frequencies from open-ended comments
monitoring_imp_comment_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(comment = opportunitiesb_for_improved_cfm_implementation_cfm_monitoring_improvement_
  drop_na(comment) |>
  unnest_tokens(word, comment) |>
  anti_join(stop_words, by = "word") |>
  mutate(word = case_match(word,
    "hfos"      ~ "More HFOs/scouts/guards",
    "hfo"       ~ "More HFOs/scouts/guards",
    "scouts"    ~ "More HFOs/scouts/guards",
    "guards"    ~ "More HFOs/scouts/guards",
    "employ"    ~ "more employment",
    "employing" ~ "more employment",
    "ppe"       ~ "PPE",
    .default = word
  )) |>
  filter(nchar(word) > 2, !word %in% c("cfm", "cfmg", "forest", "community", "people",
                                       "monitoring", "improve", "improvement")) |>
  count(word, name = "freq") |>
  filter(freq > 1)
```

```r
# Combine checkbox labels and comment words
monitoring_imp_all <- bind_rows(monitoring_imp_freq, monitoring_imp_comment_freq) |>
  group_by(word) |>
  summarise(freq = sum(freq), .groups = "drop")

set.seed(42)
ggplot(monitoring_imp_all, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 22)) +
  scale_colour_gradient(low = "#74ADCB", high = "#1B618C") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 5: Suggested improvements to CFM monitoring, as identified by community female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each suggestion.

Training needs

We asked community FGD groups to identify the types of training most needed to improve CFM in their communities (Figure 6). The responses appear to reflect the activities they are most involved with, such as fire management and conservation. However, it was interesting to see others also mentioned frequently, such as forest law and policy, agroforestry, nursery management, climate-smart agriculture and sustainable harvesting. These training needs reflect the interests of the communities and so provide a valuable insight and guidance for FD and the organisations supporting CFM in Zambia.

```
prefix_tr <- "opportunitiesb_for_improved_cfm_implementation_cfmg_training_type_"

training_labels <- c(
  "forest_conservation_and_biodiversity"                                    = "Conservation"
  "agroforestry"                                                            = "Agroforestry"
  "tree_planting_and_nursery_management"                                    = "Nursery manage
  "forest_law_and_policy"                                                   = "Forest law & p
  "governance_and_leadership"                                               = "Governance & l
  "conflict_resolution_and_community_engagement"                            = "Conflict resol
  "climate_smart_agriculture"                                               = "Climate-smart
  "livestock_management"                                                    = "Livestock mana
  "aquaculture"                                                             = "Aquaculture",
  "climate_change_adaptation_and_mitigation"                               = "Climate change
  "sustainable_forest_product_harvesting_e_g_honey_mushrooms_firewood"     = "Sustainable ha
  "financial_management_and_record_keeping"                                 = "Financial mana
  "business_development_and_entrepreneurship"                               = "Business devel
  "marketing_and_value_addition_of_forest_products"                        = "Marketing & va
  "fire_management_and_prevention"                                          = "Fire manageme
)

full_tr_vars <- paste0(prefix_tr, names(training_labels))

training_freq <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(all_of(full_tr_vars)) |>
  summarise(across(everything(), ~ sum(. == 1, na.rm = TRUE))) |>
  pivot_longer(everything(), names_to = "variable", values_to = "freq") |>
  mutate(word = training_labels[sub(prefix_tr, "", variable, fixed = TRUE)]) |>
  filter(freq > 0) |>
  select(word, freq)

set.seed(42)
ggplot(training_freq, aes(label = word, size = freq, colour = freq)) +
```

```
geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
scale_size(range = c(3, 22)) +
scale_colour_gradient(low = "#74ADCB", high = "#1B618C") +
theme_minimal(base_size = 14) +
theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 6: Training needs identified by community female, male, and youth FGDs. Word size is proportional to the number of FGDs reporting each training type.

Comments from the enumerators

So far, throughout the survey the responses have been those of the interviewees. However, after each FGD session, the enumerators recorded their general observations in an open-ended comment field. We coded the substantive comments thematically and these are shown in Figure 7, with size proportional to the number of FGD sessions for which that theme was noted. This is a very informative figure as a general take-home of the issues arising from the implementation of CFM in Zambia. Most prominently featured is 'capacity building', recognising that many communities have limited capacity and are currently struggling to implement CFM. Also, prominent were 're-engagement needed', 'CFMG dormant' and 'partner challenges', reflecting inadequate support from the forest department and other supporting organisations, 'youth marginalisation' potentially reflecting elite capture (which was also mentioned directly) and indicating that benefits may not be being shared equitably, and 'leadership challenges' suggesting the CFMG executive may not be functioning appropriately.

```
cfmg_comments <- cfmg |>
  filter(interview_type %in% c("cfmg_female", "cfmg_male", "cfmg_youth")) |>
  select(comment) |>
  drop_na(comment) |>
  filter(!str_detect(comment, regex(
    "^(none|nil|no comment|no comments|replacement|mixed|nil\\.?)$|^[A-Z][a-z]+(\\s[A-Z]?[a-z
    ignore_case = TRUE
  )))

theme_patterns <- c(
  "Youth marginalization"           = "youth.{0,30}sidelin|sidelin.{0,30}youth|youth.{0,20}no
  "Information gaps"                = "information gap|not aware of|lack.{0,15}information|in
  "Capacity building"               = "capacity build",
  "Enterprise development"          = "enterprise develop|develop.{0,15}enterprise|no.{0,10}e
  "Re-engagement needed"            = "re.engag|reengag|rengag|reorientation",
  "Benefits not realized"           = "not.{0,10}benefit|yet to.{0,20}benefit|not reaping|no
  "CFMG dormant"                    = "dormant|not been functioning|not been active|not been
  "Women marginalized"              = "women.{0,20}unfairly|women.{0,20}not allowed|women.{0
  "Elite capture"                   = "headman.{0,20}benefit|monopol|presiding.{0,10}everythi
  "Lack of transparency"            = "not transparent|information.{0,15}only.{0,10}executive
  "Leadership challenges"           = "change.{0,10}leadership|review.{0,10}constitution|cons
  "Partner challenges"              = "partner.{0,20}pulled out|partner.{0,20}not fulfil|not
  "User groups non-functional"      = "user group.{0,5}non functional|user group.{0,5}not fun
  "Carbon credit delays"            = "carbon.{0,20}not.{0,15}receiv|not.{0,15}receiv.{0,20}c
  "Traditional leader interference" = "traditional leader.{0,10}interfere|chiefdom.{0,10}succ
  "HFO challenges"                  = "hfo.{0,10}not.{0,10}equip|hfo.{0,10}no.{0,10}protectiv
  "High illiteracy"                 = "illiteracy|illiterate",
  "Market access"                   = "lack.{0,5}market|market.{0,5}challenge|no market|lack
  "CRB/CFMG overlap"               = "crb.{0,5}cfmg|cfmg.{0,5}crb|operating.{0,5}crb"
)

theme_counts <- map_dfr(names(theme_patterns), function(theme) {
  n <- sum(str_detect(cfmg_comments$comment, regex(theme_patterns[[theme]], ignore_case = TRU
  tibble(word = theme, freq = n)
}) |>
  filter(freq > 0) |>
```

```
  arrange(desc(freq))

set.seed(42)
ggplot(theme_counts, aes(label = word, size = freq, colour = freq)) +
  geom_text_wordcloud(eccentricity = 0.65, seed = 42) +
  scale_size(range = c(3, 18)) +
  scale_colour_gradient(low = "#6ABFB8", high = "#1A6B63") +
  theme_minimal(base_size = 14) +
  theme(plot.margin = margin(0, 0, 0, 0))
```

Figure 7: Key themes from enumerator observations recorded after each FGD session. Word size is proportional to the number of FGD sessions for which that theme was noted.

Discussion of the main findings

One the one hand, CFM has clearly been a positive development in the management of Zambia's forests. Over 9 Mha have been designated as CFs and the majority view across the sampled CFMGs is that CFM has had a positive effect on forest management and for livelihoods. A majority of communities reported that CFM was reducing deforestation, reducing fire frequency and increasing wildlife populations – all key indicators of SFM. Furthermore, a majority of communities recognise that CFM has brought some economic benefits though increased employment and income.

On the other hand, a proportion of CFs are clearly not doing as well. Out of the sample of 80 CFMGs five (6.25%) were defunct to the point that they could not even be included in the sample. For a number of others the CFMG exists, but is dormant and re-engagement is urgently needed. The review of governance aspects highlighted that many CFs have yet to complete all the basic legal processes and only a minority have opened bank accounts, operate receipt books, have formalised user-groups and issue permits. Similarly, few CFMGs have annual workplans and budgets, and annual reporting is rare despite being a requirement under the license. Feedback to the community from the CFMG is also clearly inadequate in many cases. As many of the benefits of CFM stem from improved governance, these issues are a serious concern and if not rectified will jeopardise the implementation of CFM across Zambia.

The FD has a statutory responsibility to support community forestry and provide oversight of CF licenses. Yet for most CFs visits from DFOs are a rarity. Furthermore, the CFMG database maintained by the FD is clearly not up to task. For example, only 40 out of the 80 sampled CFs had boundary information. Documentation of CFMG processes is often hopelessly out of date or non-existent. It may be possible for a competent DFO to maintain records for a small number of CFMGs, but at a national level it is essential that the FD makes proper use of digital tools to maintain a high quality database and dashboard. Properly implemented, this would become a tool not only for managing CFM but also for reporting on national targets. In addition, it is essential that DFOs are well trained and adequately funded to visit CFMGs and support CFM on the ground.

Overall, the feedback from communities of the impacts of CFM on SFM were quite positive. This likely reflects the devolution of the control of forests to communities, who have an interest in improving the management of their forests, and the injection of capital into forest management that has occurred mainly through the proliferation of voluntary carbon schemes. Through investment in HFOs and patrolling, fire management and alternative livelihoods, such as climate smart agriculture, the conservation NGOs and carbon enterprises supporting CFM have funded a substantial upgrade in local forest management. This makes it all the more essential that the governance aspects of CFM are improved. There is a risk that these gains may be undone unless the legal and local governance frameworks are made to function properly.

With respect to economic transformation, the picture is more mixed. About one third of CFMGs in the sample had received payouts from voluntary carbon schemes and local communities recognised the livelihood improvements brought about via increased employment, capacity development for climate smart agriculture and investment in NTFP value chains, especially honey. However, it is also evident that the employment opportunities have been somewhat limited and focused on low paying jobs, such as HFOs. Furthermore, opportunities to develop other forest product value chains have not been realised. CFM should be a form of multi-use forest management that exploits a wide variety of forest products sustainably, layering different income streams on top of one another. Zambian forests are traditionally exploited for a large number of products. For example, harvesting of food from forest in Zambia is worth between US$281-US$450 per rural household (2019 data). As this survey has shown, forest are also harvested for fuelwood, local construction timber, thatching grass, medicines, bamboo and so on. However, most of these products are harvested for household use or local markets with minimal value-addition. Many of these products have the potential to be developed into community enterprises, increasing upstream value-addition and supplying urban areas, regional and sometimes international markets. Through collaboration amoung ministries, the government of Zambia should leverage CFM for the economic development of local communities through promoting forest product enterprises.

One of the concerns of CFM is that opportunity costs for some vulnerable households may be significant, but that most of the benefits may accrue more to local elites. Interestingly, women were more supportive of CFM than other community FGD groups. Most female FGD groups also stated that women did not suffer opportunity costs disproportionally. These positive outcomes may arise because the formalisation of local forest management actually gives women a voice that they might not previously have had. In Zambia, natural resource management decisions are traditionally the purview of male community members, so CFM through formalising CFMGs, with elections and female representation, may have improved their voice in forest management. Notwithstanding these positive outcomes across most CFMGs, a minority of female FGD groups felt that opportunity costs disproportionately affected women, that women were not involved in decision making and that there were issues surrounding the transparency of benefit sharing. In some cases, community youth groups identified similar problems and community youth FGD groups tended to have a more negative perspective on CFM than the male or CFMG executive groups. It is essential that these problems are identified where they occur and addressed to ensure that CFM does not negatively affect women and youth. Much of this comes back to ensuring that the governance processes are properly implemented, but in addition it may help to enhance standardisation of benefit sharing arrangements and increase representation of women and youth on CFMGs. CFMG workplans, budgets and annual reports could also have sections dedicated to activities supporting women, youth and other vulnerable groups.

In conclusion, CFM has clearly led to a number of positive outcomes for SFM and livelihoods. Through formalising local management it has enabled investment in forest management from conservation NGOs and carbon enterprises. However, there is a substantial concern around the FD's capacity to provide the necessary oversight and support for CFM. The FD should invest in digital support tools for management of CMF, support DFOs to visit CFs regularly and provide training on CFM to forestry staff. Furthermore, the government of Zambia and private entities supporting CFM should support CFMGs to diversify forest product enterprises, increase value addition and enhance market linkages. CFM represents a significant opportunity for both improving SFM and realising economic transformation of rural communities, but it is likely to fall well short of its potential if these governance and economic development issues are not addressed.